Method selection and planning

Cohort 2- Group 1

Members

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Development Method

We went with the Agile method for this project as it was preferred in the lecture. GIven the short timeline and relatively small scope of the project, Agile methodology was the ideal fit because it allowed us to work in quick cycles rather than committing to a long-term plan from the start and allowed us to focus on iterative development and regular improvements. The main benefit of the Agile method was its flexibility, letting us adapt and pivot based on our weekly progress. This method made it easier to manage the collaboration aspect, where different team members could contribute to the same things rather than waiting for one part to finish before starting another.

Tools

File sharing and organisation - Google Drive/Docs and GitHub Repository

We chose to use Google Drive and Google Docs as our main collaborative workspace to organise and manage all the project elements. Google Drive was the most logical choice for us, as we all have university Google accounts, making it really easy to set up. This platform offers simple tools for organising files and folders on the shared drive, allowing us to keep everything from design documents to code snippets in one location that's accessible to everyone in the group.

One major benefit of Google Drive is that the documents are stored online and updated in real time, which means we can work collaboratively, even when we aren't together. This real time updating feature allows us to work simultaneously without worrying about any version control issues or having to wait for one person to finish working on a document. The ability to add comments and suggestions directly onto documents also makes it easy to give feedback, share ideas, and discuss tasks without needing to be active at the same time. This way, each team member can stay informed and up to date with the latest progress, even if they weren't involved in a specific update to a document

We decided to set up a GitHub repository to manage and streamline our code development. GitHub is a powerful tool for version control, which allows our team to work on code simultaneously while minimising conflicts and ensuring that we have a reliable history of every change made to the project.

GitHub's version control capabilities are particularly valuable because they allow us to maintain a record of every change, revert to previous versions when necessary, and debug more efficiently. If an update causes unexpected issues, we can easily go back to a previous commit to restore functionality, which gives us a safety net for experimenting and testing new features. This is crucial in collaborative coding environments, as it keeps the project moving forward without major setbacks from unintentional errors.

Additionally, GitHub serves as a centralised repository accessible to all the team members, meaning we can pull the latest updates and stay up to date with each other's progress. This is especially helpful in preventing "merge conflicts," which can arise when multiple people try to edit the same part of a project. GitHub's built-in conflict resolution tools make it easy to manage any conflicting code changes.

General chats and updates - Instagram Group Chat

For our more general informal communication we decided to simply exchange Instagram accounts and make an Instagram Group chat. This made the most sense as it is easy to use and everyone will have access to it all the time on their phones.

Discussed alternative

Our team initially considered setting up a dedicated Discord server for our main communication for the project. Discord seemed like a good option due to its popularity as a platform for collaboration, with features like voice channels, video calls, and the ability to create multiple text channels for different topics. However, after some discussion, we ultimately decided against it. One key reason was the layout and functionality of Discord, which felt less intuitive and user-friendly on mobile devices. Since many of us would need to access the server on our phones throughout the day, ease of use on mobile was a priority, and Discord's interface seemed to lack the simplicity we were looking for.

Also, we were aiming for a more informal, unstructured form of communication. While Discord servers can be organised with specific channels for different project topics (e.g., one for design, one for coding issues, etc.), we felt that too many channels could make communication feel segmented and potentially even reduce our interactions. The added structure risked creating an environment where team members might feel unsure of where to post updates or questions, leading to missed information and possibly reduced engagement from the team. We wanted a setup that felt more like a natural conversation, where people could easily share updates, discuss ideas, and ask questions without navigating a complex system of channels

Team Structure

Our team consisted of seven members, and we structured our work based on the main deliverables: documentation and implementation. This approach allowed us to divide the workload efficiently, with one group focusing on code development while the other concentrated on writing and structuring the final deliverables. We thought it would be a waste of time if all team members did the codes and the writing deliverables. We did not assign a formal leader, but we made sure responsibilities were clear for each task. We made sure not all crucial tasks solely relied on one person. So that if the task owner cannot contribute to that task again we would have another person who can continue the task. This division of labour helped us stay productive, as those working on implementation could focus on coding without having to juggle writing tasks simultaneously.

We held regular meetings twice a week. Initially, we only met on Thursday after our practical session to check progress, but we soon realised that this was not enough to keep up with our project's needs. So, we added Monday meetings to our schedule. During monday meetings, we reviewed completed work from the previous week and assigned new tasks, while Thursday meetings served as a progress check-in where we could discuss any challenges or delays. This twice-weekly rhythm helped keep everyone accountable, and it gave us enough time to troubleshoot any issues mid-week instead of waiting until our next meeting.

For communication, we started with a group chat on Instagram, but we switched to Discord. Discord offered a more organised platform, with channels that helped keep conversations about specific topics like design, testing, and documentation etc, separate. Discord's file sharing capabilities also made it easier to share updates and work together outside of our regular meeting times.

Project Plan

The main idea for our project plan is to work in smaller groups across a number of different tasks in parallel. By dividing the project into distinct tasks and assigning smaller teams to handle them, we can maximise efficiency and make progress on multiple fronts simultaneously. This structure also allows team members to specialise in areas where they feel most comfortable or skilled. These tasks and groups were decided collectively as a team to ensure everyone is not only satisfied with their responsibilities but also aware of each other's roles. This approach creates a sense of ownership and accountability, helping each team member feel invested in the project's overall success. Before assigning tasks, we thoroughly reviewed the exam document and identified its main deliverables, which allowed us to distinguish between sections that could be completed independently and those that would require collaborative effort.

One of the primary tasks is changing assessment 1 deliverables to fit the new criteria for assessment 2 and to match our group's vision instead of the previous group's. This task can be done by the whole group over the course of the entire project. This then also applies to the changes document that is one of the deliverables for assessment 2. This document will be worked on by the entire group as the person that makes changes to the assessment 1 deliverables will need to write it down with a justification.

The primary software implementation of the product has been assigned to two designated "Lead Developers," who will drive the hands-on coding and development. However, this task isn't isolated—these developers will require continuous feedback and input from the rest of the team to ensure that the implementation aligns with the outlined requirements and architectural plans. The development process will be informed by the outputs of all other tasks, making communication and integration essential for a cohesive product.

Conducting the user evaluation on a working prototype of the game. This can be carried out by 1 or more people. This is the task that requires the most communications skills as it requires the need to talk to people to get them to test the game. This also needs to have consent towards being a part of it and have knowledge on what the data will be used for.

Software testing can be done in conjunction with the development of the game. It will require communication between the pair that will do the testing and the lead developers if what is required of a function/feature is not clear in the code. This section is crucial as it is what will allow us to know if the code works properly and doesn't crash. It will also need to be done throughout the entire project as it is needed for the build of the game that will be used for user evaluation and the final build that will be handed in.

Finally, organising and overseeing all of these concurrent tasks is essential for smooth project execution. Effective communication is crucial when team members work semi-independently, as it helps prevent misunderstandings and keeps everyone aligned with the project's goals. A designated project coordinator will play a vital role in maintaining this communication flow, facilitating updates across the team, and ensuring that each task progresses as planned. This coordination not only enhances efficiency but also fosters a collaborative environment where team members feel supported and well-informed throughout the project's duration.

Key Milestones

- Get open (changeable) versions of all deliverables from the previous group. *To be completed by 05/12/24*

- Create a functional website that matches the website of the previous team and allows for our various project documents to be displayed and accessed. *To be completed by 19/12/24*

- Update assessment 1 documentation to fit the assessment 2 criteria. *To be completed by 19/12/24*

- Create a working build of the game for user evaluation. *To be completed by 28/12/24*

- Complete user evaluation on working build as previously mentioned. *To be completed by 4/1/25*

- Improve the game according to the feedback gained from user evaluation. *To be completed by 12/01/25*

- Produce deliverables for assessment 2. *To be completed by 12/01/25*